INSTITUT UNIVERSITAIRE DES SCIENCES

FACULTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TD2- ADMINISTRATION SYSTÈME

PRÉPARÉ PAR :

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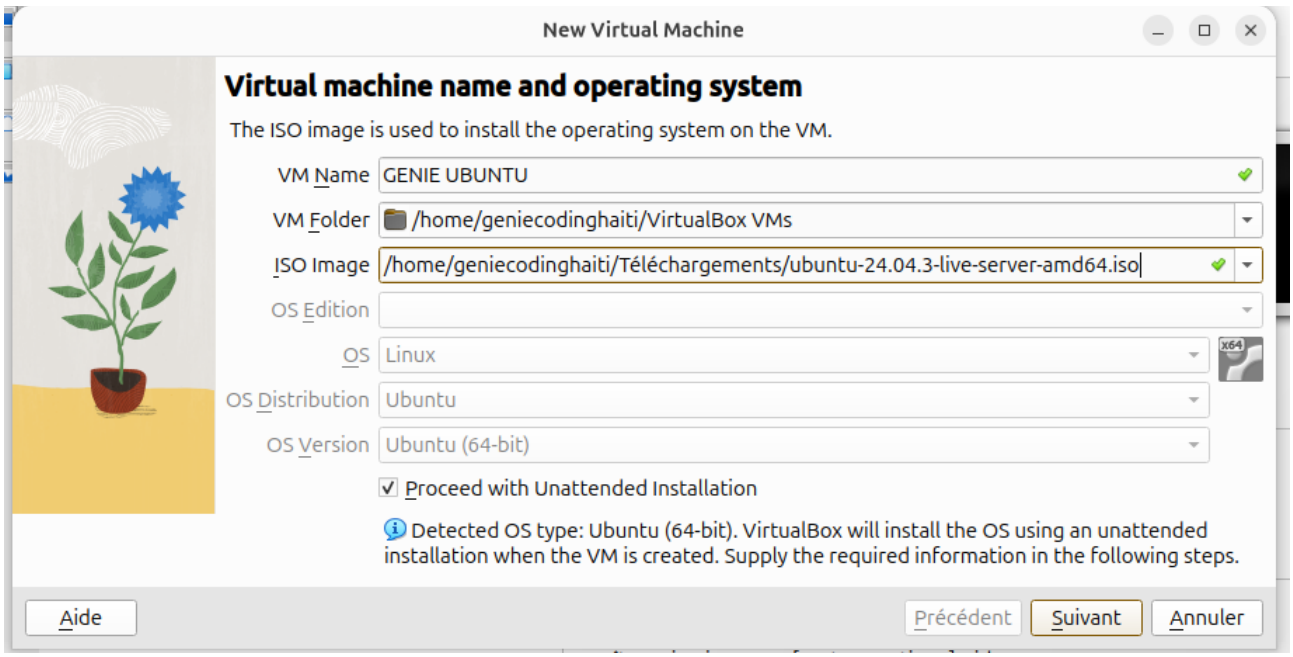
Niveau : L4-Informatique et Gestion

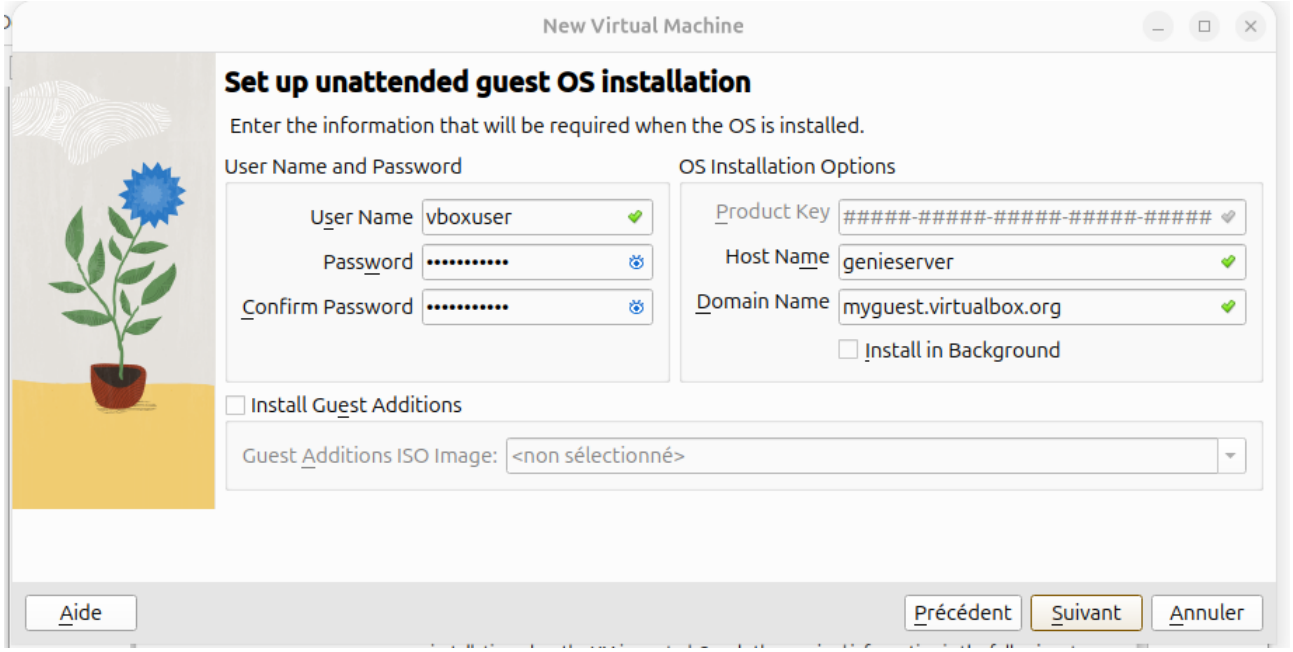
Date : 13 Novemvre 2025

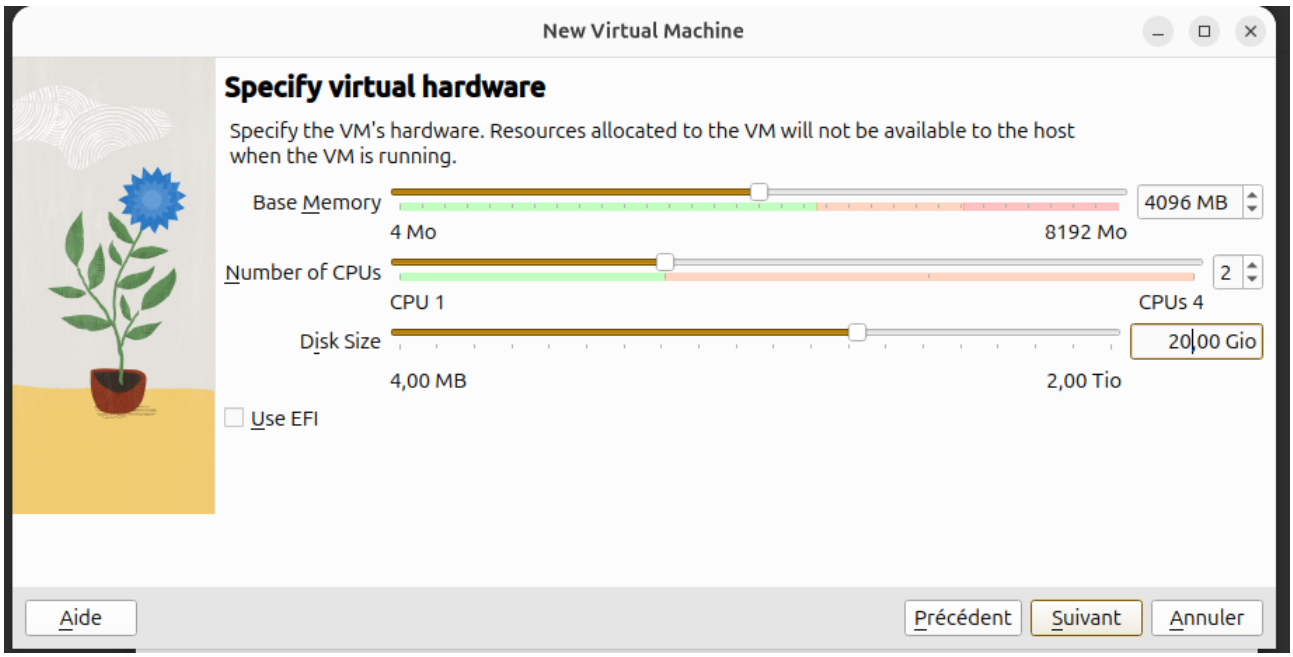
Objectif du Td :

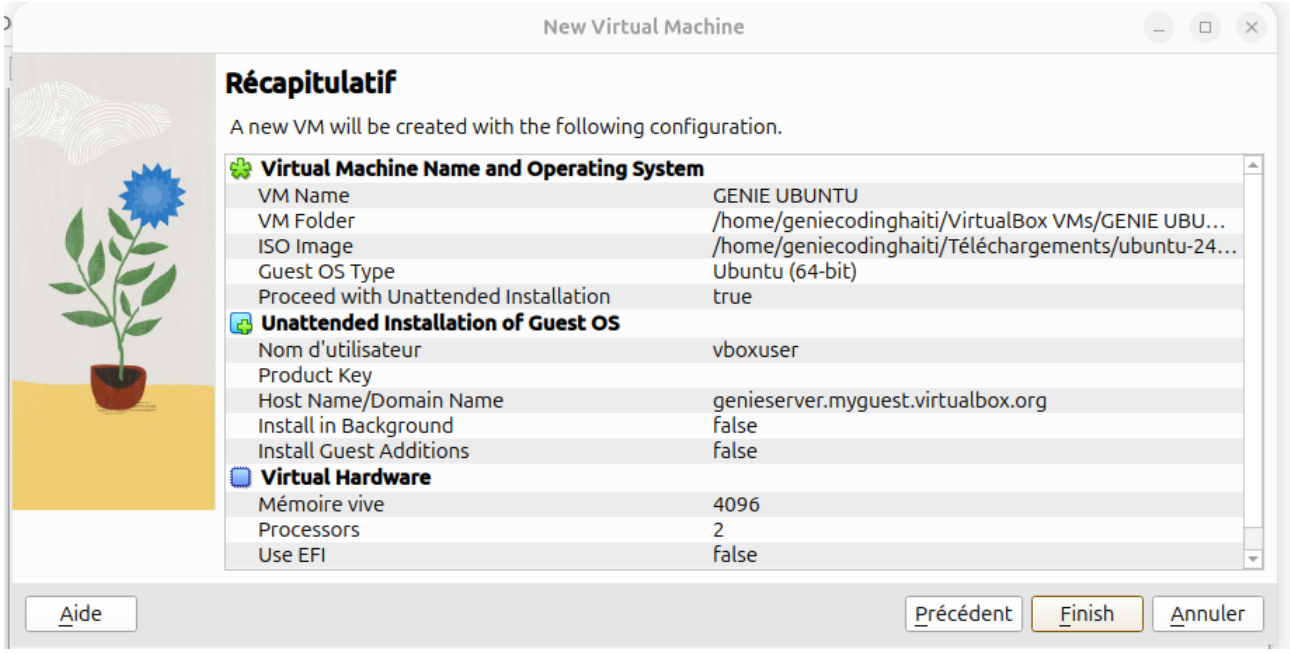
→ Ce Td vise à configurer ubuntu server sur VirtualBox et à l'installer sur cette VM.

Configuration Ubuntu Server sur VirtualBox









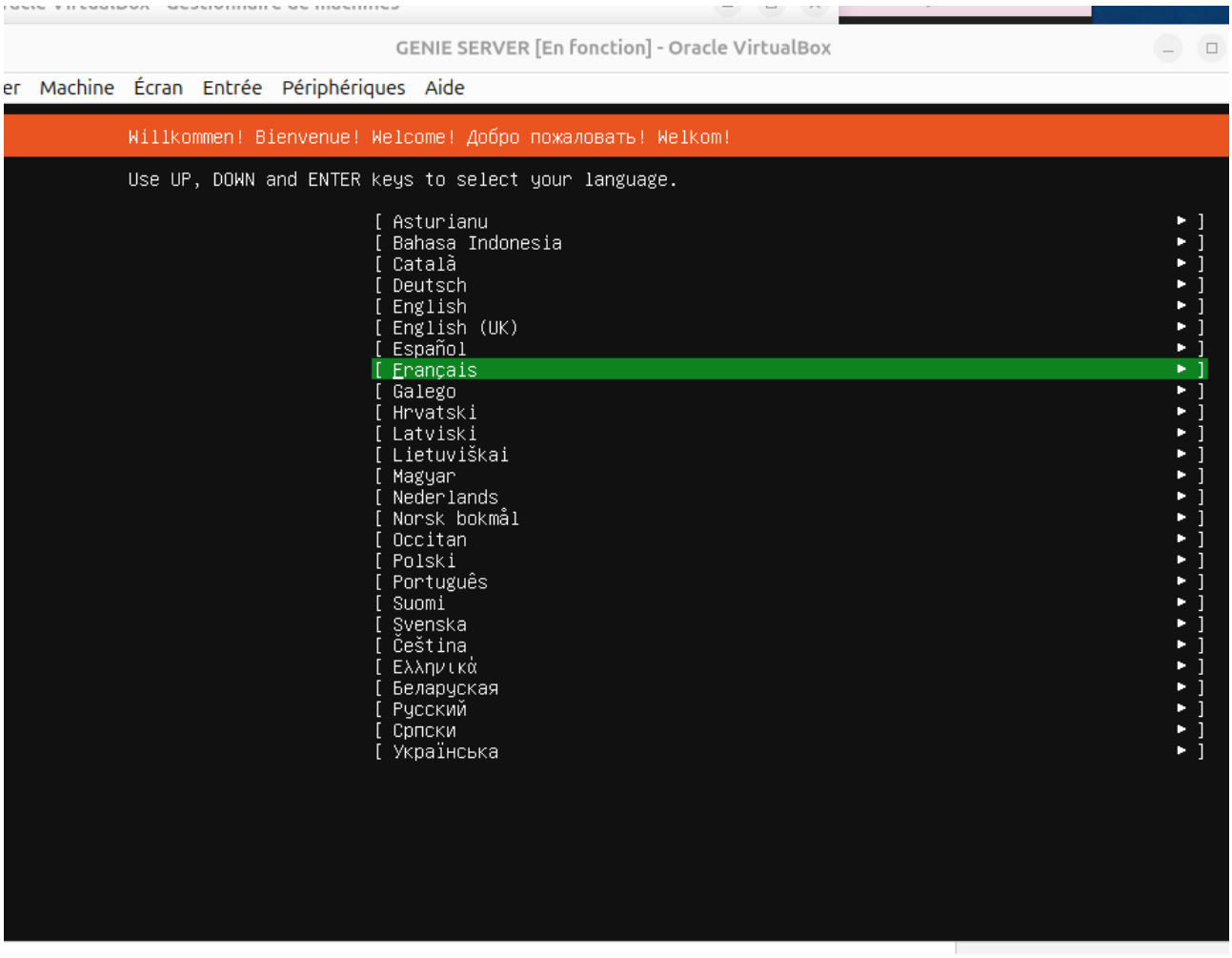
Installation de Ubuntu Server

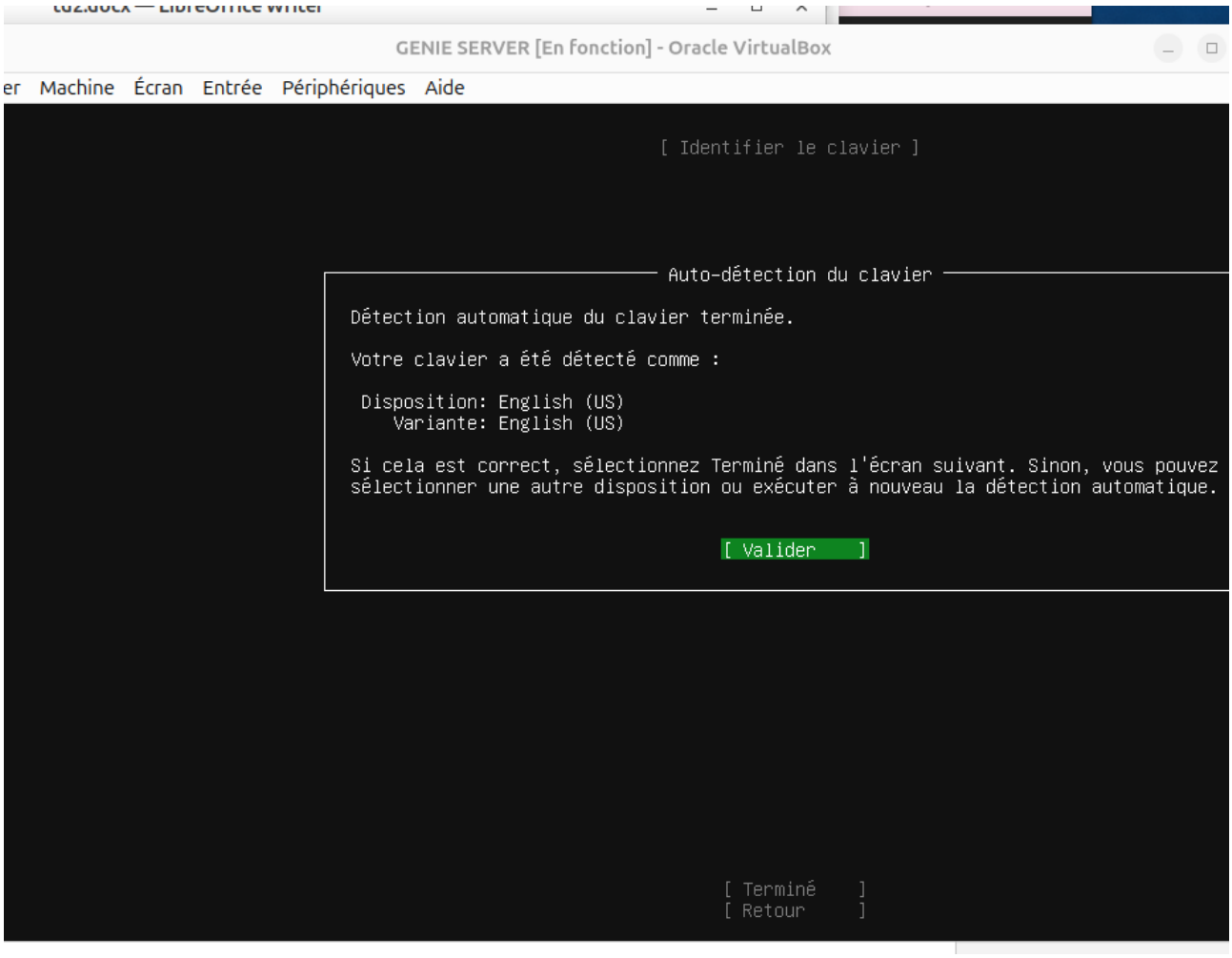
GENIE UBUNTU [En fonction] - Oracle VirtualBox

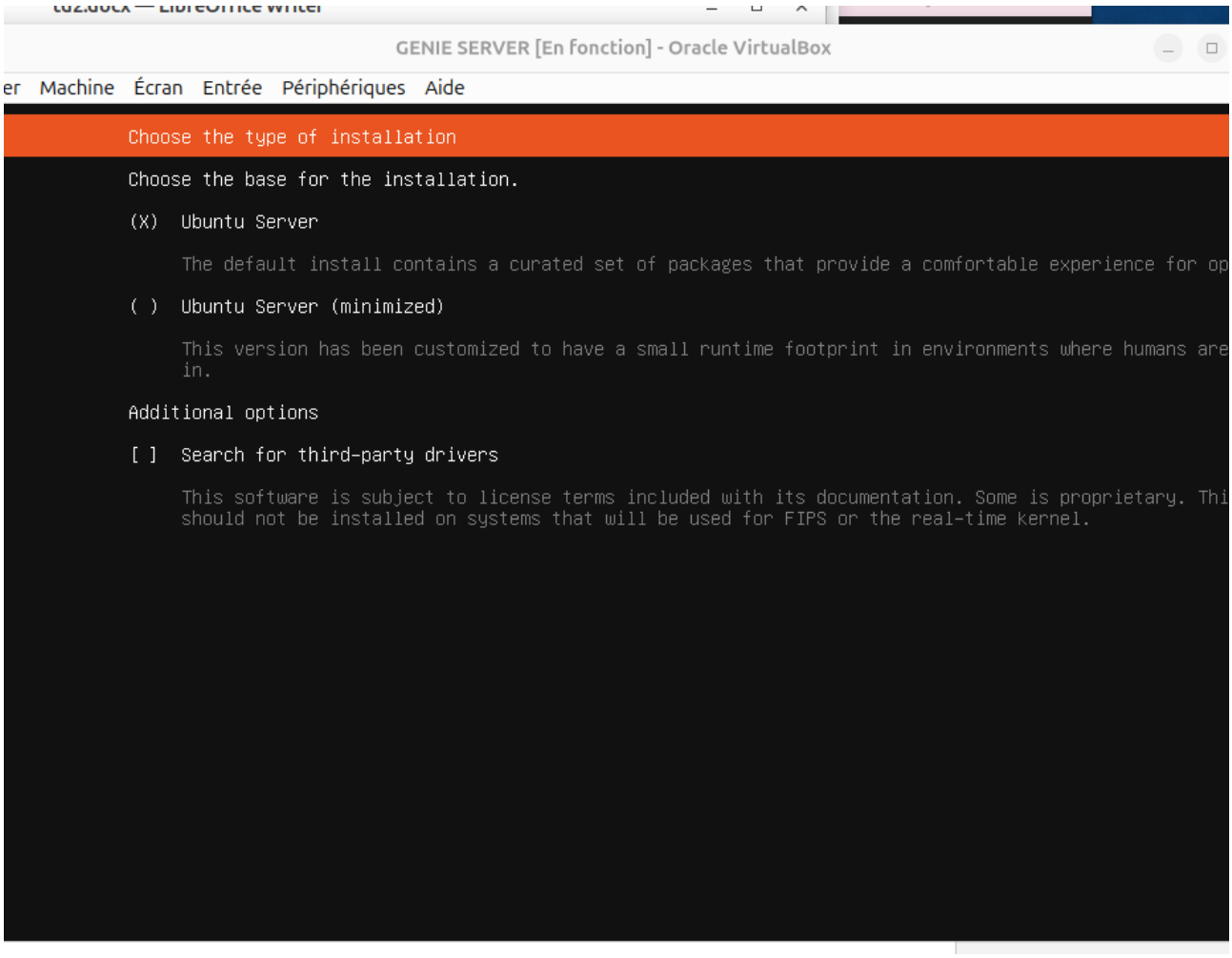
Fichier Machine Écran Entrée Périphériques Aide

```
finish: subiquity/OEM/load_metapackages_list:
finish: subiquity/OEM/apply_autoinstall_config:
start: subiquity/TimeZone/apply_autoinstall_config:
finish: subiquity/TimeZone/apply_autoinstall_config:
start: subiquity/Updates/apply_autoinstall_config:
finish: subiquity/Updates/apply_autoinstall_config:
start: subiquity/Late/apply_autoinstall_config:
finish: subiquity/Late/apply_autoinstall_config:
start: subiquity/Shutdown/apply_autoinstall_config:
finish: subiquity/Shutdown/apply_autoinstall_config:
finish: subiquity/apply_autoinstall_config:
start:  subiquity/Install/install/configure_apt: configuring apt
start: subiquity/Meta/status_GET:
start: subiquity/Meta/status_GET:
start: subiquity/Mirror/cmd-apt-config: curtin command apt-config
finish: subiquity/Mirror/cmd-apt-config: curtin command apt-config
start:   subiquity/Install/install/configure_apt/cmd-in-target: curtin command in-target
finish:   subiquity/Install/install/configure_apt/cmd-in-target: curtin command in-target
finish:  subiquity/Install/install/configure_apt: configuring apt
start:  subiquity/Install/install/curtin_install: installing system
start:   subiquity/Install/install/curtin_install/run_curtin_step: executing curtin install initial step
finish:  subiquity/Drivers/_list_drivers/wait_apt:
finish:   subiquity/Install/install/curtin_install/run_curtin_step: executing curtin install initial step
start:   subiquity/Install/install/curtin_install/run_curtin_step: executing curtin install partitioning step
start:  subiquity/Drivers/_list_drivers/cmd-in-target: curtin command in-target
start:    subiquity/Install/install/curtin_install/run_curtin_step/cmd-install: curtin command install
start:     subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning: configuring st
start:      subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin: runni
start:       subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd-b
start:        subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd-
orage devices
finish:  subiquity/Drivers/_list_drivers/cmd-in-target: curtin command in-target
finish: subiquity/Drivers/_list_drivers:
finish:        subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd
torage devices
start:        subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd-
finish:        subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd
start:       subiquity/Install/install/curtin_install/run_curtin_step/cmd-install/stage-partitioning/builtin/cmd-b
```

Ctrl droite


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!
Use UP, DOWN and ENTER keys to select your language.
[Asturianu]
[Bahasa Indonesia]
[Català]
[Deutsch]
[English]
[English (UK)]
[Español]
[Français]
[Galego]
[Hrvatski]
[Latviski]
[Lietuviškai]
[Magyar]
[Nederlands]
[Norsk bokmål]
[Occitan]
[Polski]
[Português]
[Suomi]
[Svenska]
[Čeština]
[Ελληνικά]
[Беларуская]
[Русский]
[Српски]
[Українська]


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
[Identifier le clavier]
Auto-détection du clavier
Détection automatique du clavier terminée.
Votre clavier a été détecté comme :
Disposition: English (US)
Variante: English (US)
Si cela est correct, sélectionnez Terminé dans l'écran suivant. Sinon, vous pouvez sélectionner une autre disposition ou exécuter à nouveau la détection automatique.
[Valider]
[Terminé]
[Retour]


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
Choose the type of installation
Choose the base for the installation.
(X) Ubuntu Server
The default install contains a curated set of packages that provide a comfortable experience for op
( ) Ubuntu Server (minimized)
This version has been customized to have a small runtime footprint in environments where humans are
in.
Additional options
[ ] Search for third-party drivers
This software is subject to license terms included with its documentation. Some is proprietary. Thi
should not be installed on systems that will be used for FIPS or the real-time kernel.

GENIE SERVER [En fonction] - Oracle VirtualBox

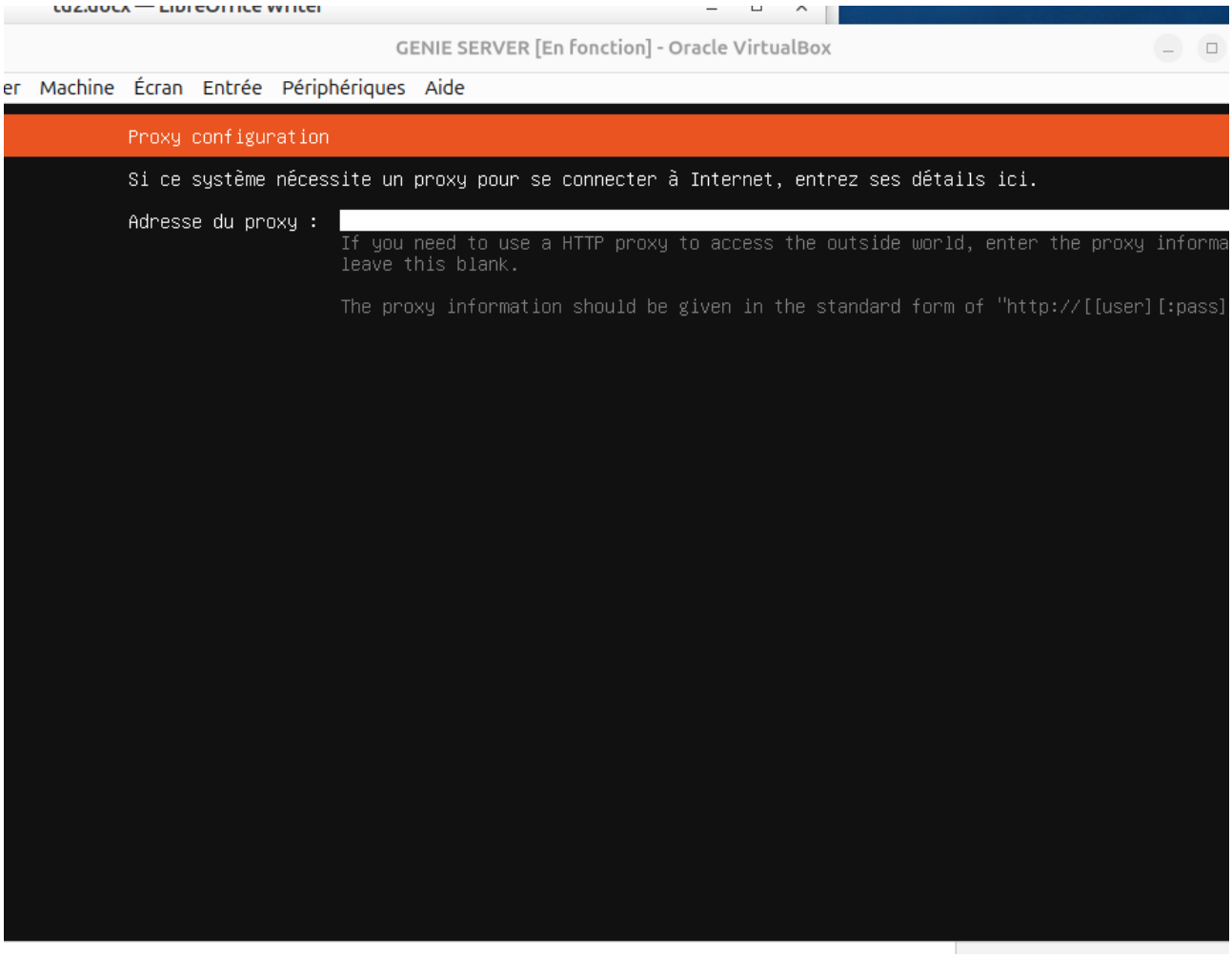
er Machine Écran Entrée Périphériques Aide

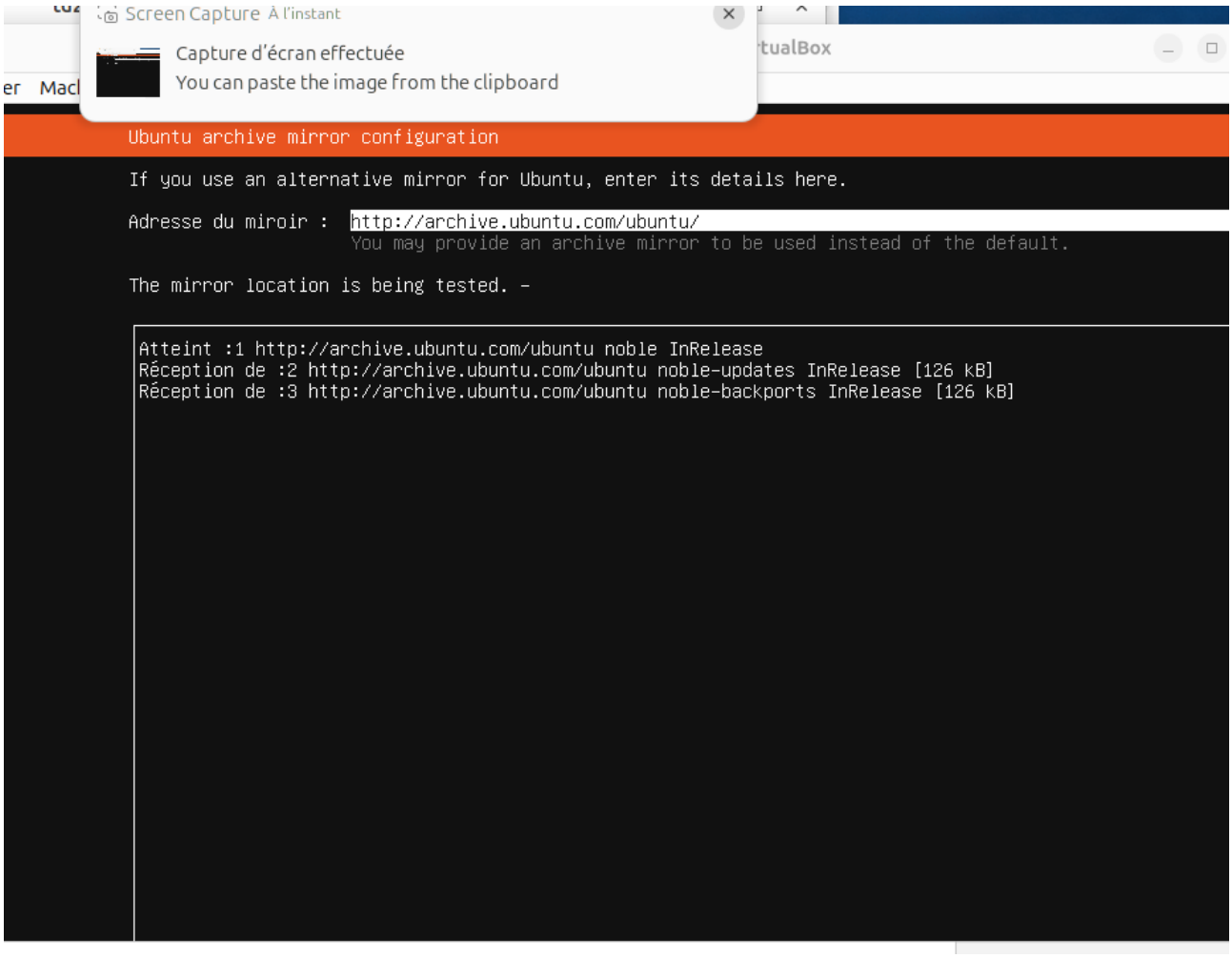
```
Network configuration

Configurez au moins une interface pour que ce serveur puisse communiquer avec les autres machines sur le
préférablement un réseau avec accès aux mises à jour.

  NAME    TYPE  NOTES
[ enp0s3  eth   -                ▸ ]
  DHCPv4  10.0.2.15/24
  08:00:27:b3:d2:5c / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapt

[ Create bond ▸ ]
```


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
Proxy configuration
Si ce système nécessite un proxy pour se connecter à Internet, entrez ses détails ici.
Adresse du proxy :
If you need to use a HTTP proxy to access the outside world, enter the proxy informa
leave this blank.
The proxy information should be given in the standard form of "http://[[user][:pass]


Screen Capture À l'instant
Capture d'écran effectuée
You can paste the image from the clipboard
tualBox
Ubuntu archive mirror configuration
If you use an alternative mirror for Ubuntu, enter its details here.
Adresse du miroir : http://archive.ubuntu.com/ubuntu/
You may provide an archive mirror to be used instead of the default.
The mirror location is being tested. -
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Réception de :3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]

GENIE SERVER [En fonction] - Oracle VirtualBox

er Machine Écran Entrée Périphériques Aide

```
Configuration de stockage guidée

Configure a guided storage layout, or create a custom one:

(X)  Utiliser un disque entier

     [ VBOX_HARDDISK_VBb3ca68d8-bfc75219                local disk 20.000G ▼ ]

     [X]  Set up this disk as an LVM group

          [ ]  Encrypt the LVM group with LUKS

                                Phrase de passe :

                            Confirmez la phrase de passe :

                                                   [ ]  Also create a recovery key
                                                        The key will be stored as ~/recovery-key.txt in the live
                                                        copied to /var/log/installer/ in the target system.

( )  Custom storage layout
```

GENIE SERVER [En fonction] - Oracle VirtualBox

er Machine Écran Entrée Périphériques Aide

```
Configuration du stockage

SOMMAIRE DU SYSTÈME DE FICHIERS

  POINT DE MONTAGE    TAILLE   TYPE      TYPE DE PÉRIPHÉRIQUE
[ /                   10.000G  new ext4  nouveau LVM logical volume        ► ]
[ /boot                1.771G  new ext4  nouveau partition de disque local ► ]


DISQUES DISPONIBLES

  PÉRIPHÉRIQUE                                    TYPE                TAILLE
[ ubuntu-vg (nouveau)                             LVM volume group    18.222G  ► ]
  espace libre                                                         8.222G  ►

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]


PÉRIPHÉRIQUES UTILISÉS

  PÉRIPHÉRIQUE                                    TYPE                TAILLE
[ ubuntu-vg (nouveau)                             LVM volume group    18.222G  ► ]
  ubuntu-lv     nouveau, to be formatted as ext4, mounted at /        10.000G  ►

[ VBOX_HARDDISK_VBb3ca68d8-bfc75219               disque local        20.000G  ► ]
  partition 1   nouveau, BIOS grub spacer                                1.000M  ►
  partition 2   nouveau, to be formatted as ext4, mounted at /boot       1.771G  ►
  partition 3   nouveau, PV of LVM volume group ubuntu-vg              18.225G  ►
```

GENIE SERVER [En fonction] - Oracle VirtualBox

er Machine Écran Entrée Périphériques Aide

Profile configuration

Enter the username and password you will use to log in to the system. You can configure SSH access on a
password is still needed for sudo.

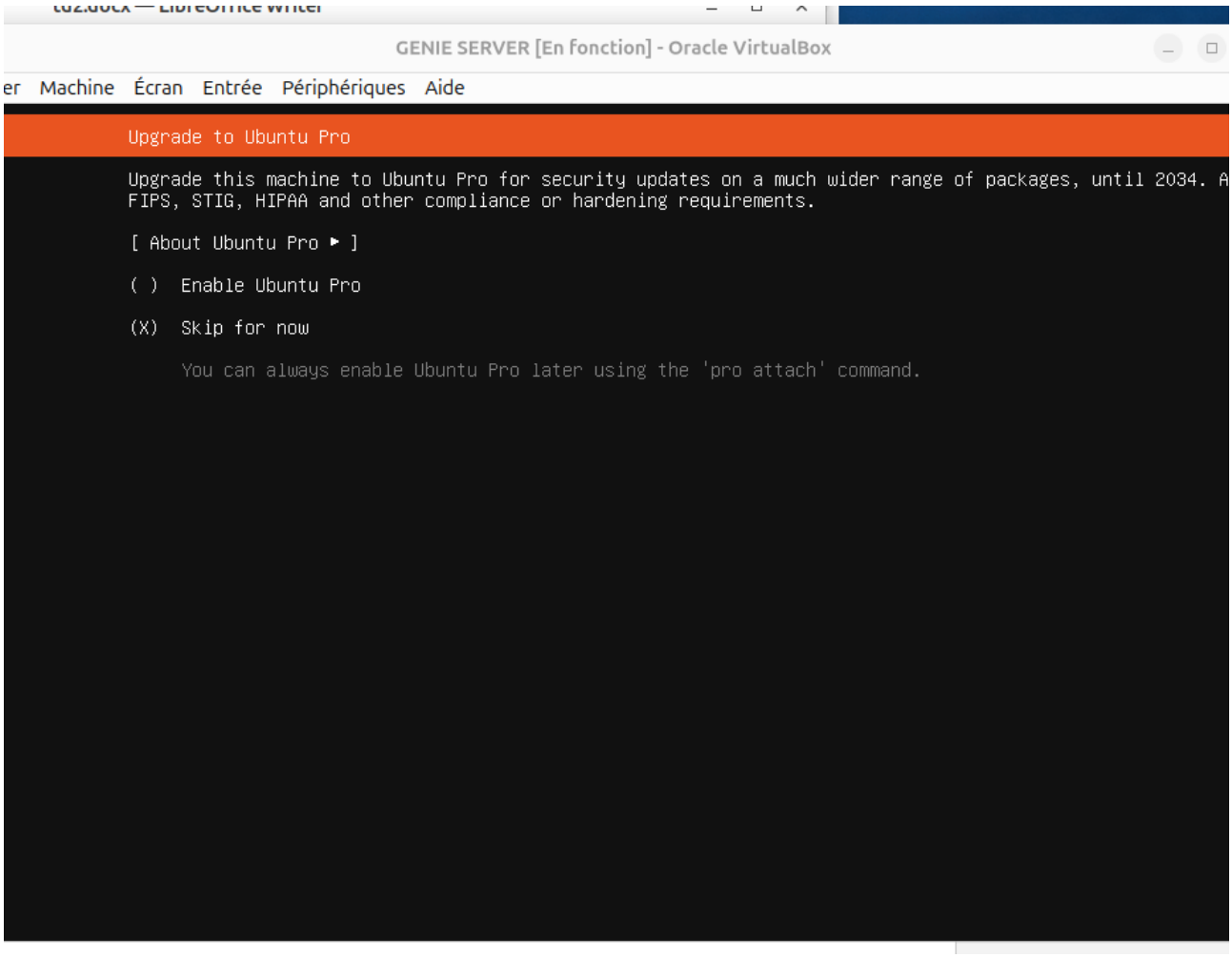
Votre nom : PIERRE Yann Lelay

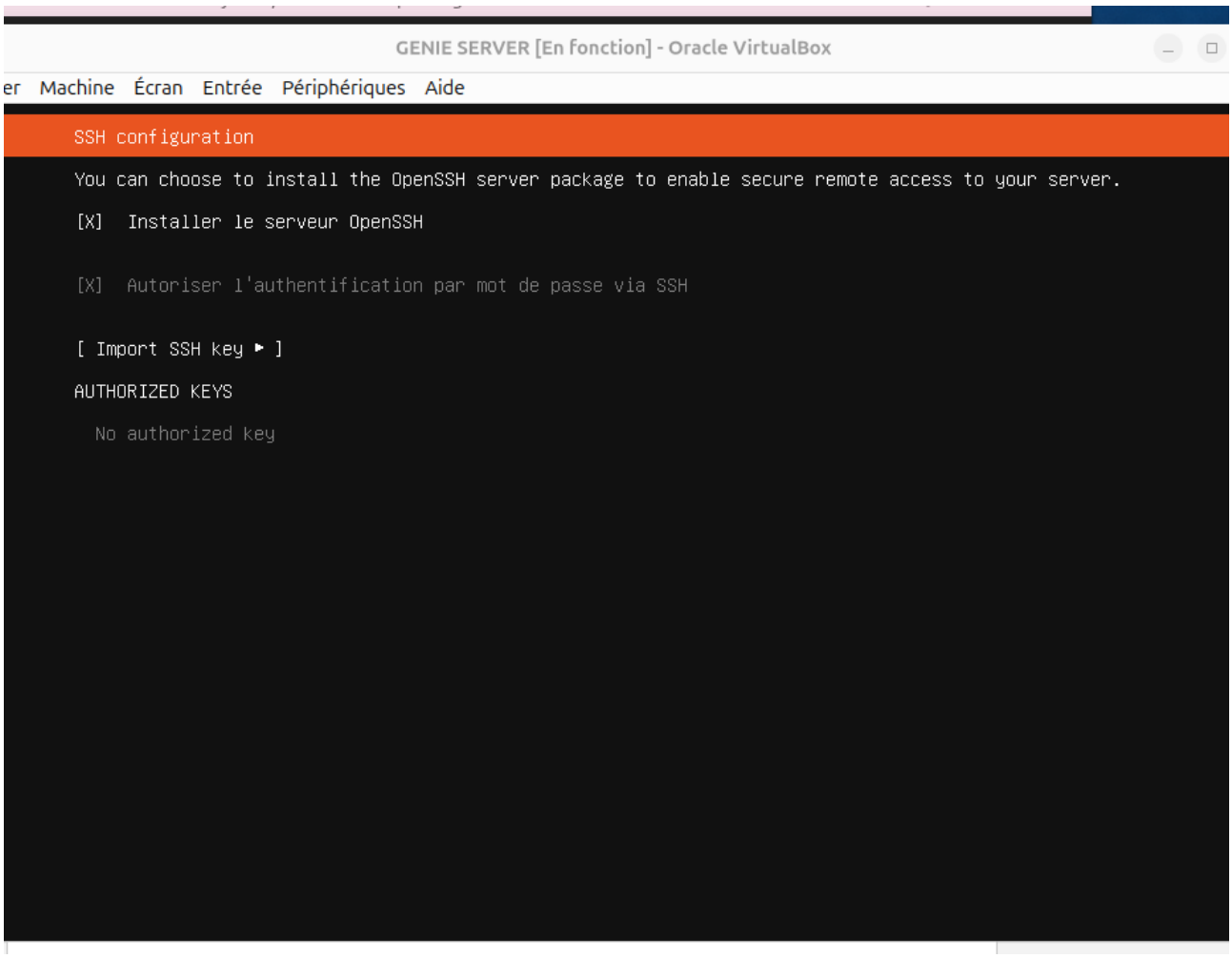
Your servers name: genie_server
The name it uses when it talks to other computers.

Choisir un nom d'utilisateur : legenie

Choisir un mot de passe : *******

Confirmer votre mot de passe: *******


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
Upgrade to Ubuntu Pro
Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2034. A
FIPS, STIG, HIPAA and other compliance or hardening requirements.
[About Ubuntu Pro ▸]
( ) Enable Ubuntu Pro
(X) Skip for now
You can always enable Ubuntu Pro later using the 'pro attach' command.


GENIE SERVER [En fonction] - Oracle VirtualBox
er Machine Écran Entrée Périphériques Aide
SSH configuration
You can choose to install the OpenSSH server package to enable secure remote access to your server.
[X] Installer le serveur OpenSSH
[X] Autoriser l'authentification par mot de passe via SSH
[Import SSH key ►]
AUTHORIZED KEYS
No authorized key

GENIE SERVER [En fonction] - Oracle VirtualBox

er Machine Écran Entrée Périphériques Aide

```
Featured server snaps

These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more detail
publisher and versions available.

[ ] microk8s            canonical✓          Kubernetes for workstations and appliances
[ ] nextcloud           nextcloud✓          Nextcloud Server - A safe home for all your data
[ ] wekan               xet7                Open-Source kanban
[ ] canonical-livepatch canonical✓          Canonical Livepatch Client
[*] rocketchat-server   rocketchat✓         Rocket.Chat server
[ ] mosquitto           mosquitto✓          Eclipse Mosquitto MQTT broker
[ ] etcd                canonical✓          Resilient key-value store by CoreOS
[ ] powershell          canonical✓          PowerShell for every system!
[ ] sabnzbd             safihre             SABnzbd
[ ] wormhole            snapcrafters        get things from one computer to another, safely
[*] aws-cli             aws✓                Universal Command Line Interface for Amazon Web Services
[*] google-cloud-sdk    google-cloud-sdk✓   Google Cloud SDK
[ ] slcli               softlayer           Python based SoftLayer API Tool.
[ ] doctl               digitalocean✓       The official DigitalOcean command line interface
[ ] keepalived          keepalived-project✓ High availability VRRP/BFD and load-balancing for Linux
[*] prometheus          canonical✓          The Prometheus monitoring system and time series database
[ ] lxd                 canonical✓          LXD - container and VM manager
```

GENIE SERVER [En fonction] - Oracle VirtualBox

er Machine Écran Entrée Périphériques Aide

```
executing curtin install partitioning step
 curtin command install
  configuring storage
   running 'curtin block-meta simple'
    curtin command block-meta
     removing previous storage devices
     configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring format: format-0
    configuring partition: partition-2
    configuring lvm_volgroup: lvm_volgroup-0
    configuring lvm_partition: lvm_partition-0
    configuring format: format-1
    configuring mount: mount-1
    configuring mount: mount-0
executing curtin install extract step
 curtin command install
  writing install sources to disk
   running 'curtin extract'
    curtin command extract
     acquiring and extracting image from cp:///tmp/tmpfaj2envx/mount
configuring keyboard
 curtin command in-target
executing curtin install curthooks step
 curtin command install
  configuring installed system
   running 'curtin curthooks'
    curtin command curthooks
     configuring apt configuring apt
     installing missing packages
     Installing packages on target system: ['grub-pc']
     configuring iscsi service
     configuring raid (mdadm) service
     configuring NVMe over TCP
     installing kernel /
```

[View full log]

```
legenie@genieserver:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
legenie@genieserver:~$ _
```

GENIE SERVER [En fonction] - Oracle VirtualB

Fichier Machine Écran Entrée Périphériques Aide

```
legenie@genieserver:~$ ping -c 4 google.com
PING google.com (142.251.35.238) 56(84) bytes of data.
64 bytes from tzmiaa-aa-in-f14.1e100.net (142.251.35.238): icmp_seq=1 ttl=255 time=73.4 ms
64 bytes from tzmiaa-aa-in-f14.1e100.net (142.251.35.238): icmp_seq=2 ttl=255 time=161 ms
64 bytes from tzmiaa-aa-in-f14.1e100.net (142.251.35.238): icmp_seq=3 ttl=255 time=62.0 ms
64 bytes from tzmiaa-aa-in-f14.1e100.net (142.251.35.238): icmp_seq=4 ttl=255 time=140 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 62.002/109.119/160.896/42.262 ms
legenie@genieserver:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              392M  1,1M  391M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv  9,8G  4,6G  4,7G  50% /
tmpfs                              2,0G     0  2,0G   0% /dev/shm
tmpfs                              5,0M     0  5,0M   0% /run/lock
/dev/sda2                          1,8G  100M  1,6G   7% /boot
tmpfs                              392M   12K  392M   1% /run/user/1000
legenie@genieserver:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3,8Gi       1,2Gi       454Mi       1,1Mi       2,4Gi       2,6Gi
Swap:             0B          0B          0B
legenie@genieserver:~$ sudo systemctl status ssh
[sudo] password for legenie:
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
legenie@genieserver:~$ _
```

Conclusion :

En faisant ce TD, j'arrive par comprendre comment configurer Virtual Box pour l'installation d'ubuntu server et de comprendre etapes par etapes chaque interface lors de l'installation… Après l'installation me mettre à l'avec l'environnement Shell d'Ubuntu server en me servant des commandes pour tout ce que je veux faire , sans interface graphique.